

Lec 19 Intuition behind KL Divergence - Part A

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Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and the large language models. Now we are going to start with week three. In the previous session we had a seen autoencoders and in this particular week we'll be starting with variational autoencoders and more specifically we'll be looking into the fundamental concepts. So the following are the contents for the week three. We'll be first seeing what is the intuition behind the KL divergence. Then we will look into the VAE architecture.

[00:34] Under VAE architecture we'll be looking into three important components. Encoder, latent space, decoder. Then we will look into evidence lower bound which is elbow. Then we will see variational autoencoder loss function. Under this loss function there are two components. One is reconstruction loss, the second is KL divergence. Then we will be looking into reparameterization trick. Towards the end of this particular week we will solve one numerical example so that the complete working mechanism of variational autoencoder will be clear.

[01:11] Kindly note in this particular week there will not be any hands-on. So but the hands-on on VAE will be done in the week four. So in the week four first session will be

the hands-on implementation of VAE. So this is this particular today's session content. We will look into the intuition behind the KL divergence. In this more specifically we will divide this KL divergence concept into two sessions. In this particular video we will be looking into the basic concepts of KL divergence. We look into two directions, something called as forward and reverse KL divergence. And there is two behavior called a mode covering and mode seeking.

[01:57] So the following are the contents which we'll be covering in this particular session. And in the next session, we will see something called as self-information, entropy, then the cross-entropy. And we will also look into a detailed derivation of KL divergence. So, with this, let's get uh started with the today's session. If you look into, there are two points. So, P and Q. And these two points are in two-dimensional space. Now, if you want to find out a distance between these two points, it is very simple. Measuring the distance between two points which are in two-dimensional space is very simple, and we will be using an Euclidean distance. One of the measure can be an Euclidean distance. Now, instead of two

[02:46] points, instead of the P and Q as a point, what if they are a probability distribution? What do you mean by probability distribution? Now, P and Q are no more the points. They are actually the probability distribution. Now, we want to measure how different two probability distributions are. How is P of X different from Q of X or how Q of X is different from P of X? We want to measure how these two probability distributions are different. Now, if at all if it was a points, we measured the distance using Euclidean distance. But now, they are a probability distribution. Now, if you want to measure how far they are, then we will be using something called as KL divergence, and we call it as Kullback-Leibler

[03:38] divergence. So, in short, we'll be calling it as a KL divergence. So, I hope you have understood why do we require KL divergence. Now, when I say probability distribution, we can have either discrete or we can have this is discrete and this is a continuous. Either you can you know, KL divergence can be applied if you want to measure two probability distributions. Now, that probability distribution can be either a discrete distribution or it can be a continuous distribution. So, this is all the introduction why we require a KL divergence.

[04:16] Now, with this, let us see what exactly is the formal definition of a KL divergence. So, KL divergence is mainly used to measure how one probability distribution Q of X differs from a reference probability distribution P of X. Sometime back we said we require two probability distributions. Those two probability distributions are P of X and Q of X. In this P of X we are considering it as a reference probability distribution.

[04:50] Therefore, KL divergence measures how one probability distribution Q of X differs from a reference probability distribution P of X. I hope the difference I mean definition is clear to all of you. Now, what all possible cases we can imagine? Both

these two probability distributions are exactly identical with each other. Probably we can have something like this. If you see there are two probability distributions P of X and Q of X . So, here if you observe very carefully, P of X is denoted with blue and Q of X is red. And if you look into this particular case, they both are identical. That means P of X is exactly equivalent to Q of X .

[05:41] » [snorts] » In that case, is it not there is a perfect overlap or a complete overlap? If that is the case, what will be the KL divergence value? It will be zero. So, it is all about measuring how these two distributions are. So, because both are exactly identical, the KL divergence value is zero. What is the second possible case? We can have a second case as something like this. Here, what exactly is happening? There is a good overlap, but not a complete overlap.

[06:13] There is a difference between this and this, right? So, there is you know, when you look into this, what will be the KL divergence value here? The KL divergence value will be very small. Why KL divergence value will be very small? Because the two distributions are They have a very nearby means, and there is a good overlap. Because of this, the KL divergence value will be very small. And what is the third case? Can you imagine what will be the third case and how do that image looks?

[06:46] This is the third case, where the two distributions are completely far from each other. So, then what will be the KL divergence value? Definitely, the value will be very large here. So, if you see, there is a very little overlap. When there is a little overlap, definitely the the mean is very far. So, therefore, the KL divergence value will be very large. So, from this slide, what is that we have to understand? We have to understand the definition and what exactly is KL divergence. It is used to measure how a probability distribution Q of X differs from a reference probability distribution P of X .

[07:28] And the following are the three cases. One is identical, second is a slight different, third is very much different. When it is identical, KL divergence value will be zero. When there is a small difference, KL divergence is small. When there is a huge difference, KL divergence value will be very large. Okay. Now, we are getting into one of the most important concept here, and this is called the KL divergence. The It can be measured in two directions. Now, the very first direction is a forward KL divergence.

[08:05] Understand this very carefully. This is very important. Now, KL divergence uh we have two types. One is a forward KL. So, in the next slide, you will be seeing something called as the reverse KL. Now, we have also seen that distributions can be either a discrete distribution or it can be a continuous distribution, right? Suppose, if it is a discrete distribution, how is that forward KL divergence is written? So, this is a formula for forward KL divergence. And here, we know that You know, when we are measuring, you need to have two probabilities uh distributions. Those two probability

distributions are P and Q .

[08:48] And what we are discussing here is the forward KL. So, how is that you will be measuring the forward KL? The formula used for forward KL is this. It is summation of P of X into $\log$ of P of X divided by Q of X . Don't worry. We will be seeing how did we get this derivation in our next session. In this particular session, we are only focusing on all the terms and its definition. So, what we are now seeing is a forward KL. And we have also seen it can be used for both discrete as well as continuous. Now, what we are seeing now is for a discrete, how is that the forward KL divergence for What is the forward KL divergence formula? This is the formula.

[09:34] Now, we have still not seen the definition. We will see the definition after we look into a diagram. Now, what exactly is a forward KL divergence? Look into this. We require two probabilities distributions. One is P of X , second is Q of X . Now, when you look into P of X is denoted with a black and the Q of X is denoted with the red so that you can easily identify it. Now, when you look into this entire distribution, more specifically, look into this part. What exactly is happening here?

[10:06] Is it not there exists some distribution, but Q did not cover that? Q is not there. P has some value, but the Q is not being covered. So, can you can we understand that from this diagram? Here, P of X is greater than zero and the Q of X value is zero or closer to zero. In this, we have taken exactly zero here. What is the meaning of this? In this particular scenario, what exactly is forward KL divergence? It strongly penalizes Q .

[10:41] Why? For not covering the region where P has the probability mass. Correct? So, can we now understand with this? Probably, the learned distribution did not cover that part of data. Can we imagine in this way? Right? So, this is what is a forward KL divergence where it is strongly penalizes Q when it fails to cover the region where P has the probability mass. So, in this case, what exactly is going to happen? Here, Q value is zero.

[11:15] Something divided by zero becomes an infinity. Then, definitely KL forward KL value will be very large. Right? So, then there is something wrong. Then, model will do model will take an appropriate action. What it will take, let us see that. Now, we have also seen that KL divergence can also be applied for a continuous. So, if it is a continuous probability distribution, then what is the formula used for? The formula is this. We have an integration. We have an integral P of X $\log$ P of X divided by Q of X DX . So, this is what is the formula for continuous probability distribution under the forward KL.

[12:01] How is that the diagram looks like? Is it not this is P of X and this is the second distribution Q of X ? Is it not we have two distributions now P of X and Q of X ? Now, when you overlap it, when you overlap, you observe here this is the region where the model did not captured, which means there is a some data here. There is

a mass. P has a mass, but Q did not capture it. This is the forward KL. Now, if P has some value and Q has not covered it, then what exactly KL divergence does? It gives a heavy penalty to Q for missing the regions of P.

[12:48] Okay, I hope you have understood this. If this is still not clear, going forward we are seeing some more examples. Now, if this is the case, what is that the measure the model will take? So, forward the KL, what it will do? It pushes Q to spread out and cover the entire support of P. What exactly is the meaning of this? Look at this particular diagram. Now, if you look into this, Q has not covered this region at all, right? Now, what is the solution to this? The same can you look into this?

[13:21] It is a multimodal where you have a two modes. Now, when there are two modes, if a Q has not covered, if you look into this a diagram, in this diagram Q has not covered one more region at all. So, what is that forward KL divergence does? It is going to push the Q. When you are going to push the Q, is it not you are having a broader Q? When you have a broader Q, is it not it is a covering both modes of P? Uh is this clear? So, it is very simple.

[13:53] What is KL divergence? It is used to measure how far one distribution is from another. Now, you have a forward as well as a reverse. What is a forward? Forward says here Q has not covered the region where P has some mass. Then forward KL is going to penalize the Q for not covering the region where P is having some mass. So, look at this diagram. This is for discrete and this is for continuous. In continuous, P of X and Q of X are two probabilities distribution.

[14:34] Now, when you overlap it, this portion uh Q is not at all having any value. Whereas, P has a value, but Q is not having a value. This is what measure This measuring is forward KL. Now, what is that forward KL ensures? Now, Q has not covered it, right? This portion Q has not covered. If Q has not covered, what is one way that you can make Q to cover? Is it not you will hold Q and then you will spread you will expand it. When you expand, is it not it looks something in this way?

[15:10] Now, here is it not that it is covering a both? But, when you look into, there is a some other drawback also. That we will see shortly. So, I hope with the definition of a forward KL is clear to you. Now, let us see the next direction called reverse KL. Now, what is a reverse KL? Reverse KL is denoted in this way. It is a $Q \parallel P$. And what is the formula? It is $\sum Q(x) \log \frac{Q(x)}{P(x)}$. This is the formula for reverse KL. Now, what is the reverse KL says?

[15:49] And this is for a discrete. Why discrete? Because this is the discrete graph. Now, when you look into this, here P is having a value, Q is also having a value. But as you move towards this, look at what exactly has happened. Q has a spread into some region where P is not at all having any value. But Q has captured even that part also. So, if the learned uh distribution Q of X spreads into a region where the true distribution P of X has no mass or very little mass, then measuring this is your

reverse KL.

[16:34] Now, here what exactly is happening? Q has a spread into a region where P is not at all having any value. What is the P of X here? It is a true data you know, given data underlying distribution. So, if Q has captured something, which means that is not a part of original data distribution. I hope this is clear. You are getting this. What is the reverse KL? The reverse KL here is the learned distribution Q of X has a spread into a region where P is not at all having a value or not at all having any mass there or having very little mass. Then measuring this is called reverse KL.

[17:19] This is a for discrete. Um so, if it is a continuous, how does it look? For a continuous, the formula is integral. So, $\int Q \log Q / P \, dX$. Now, uh if you have to look into this a more mathematically, what exactly is happening? Here, P of X value is zero or closer to zero. If that is the case, is it not um now the uh reverse KL value has become very large? Because this value, if it is a zero, definitely it is infinity. If it is a closer to zero, then definitely the reverse KL value is very large.

[18:03] So, measuring what is the reverse KL is all about uh punishing a Q for placing the probability where P has little or no mass. This is a for a single modality or single mode. What if you have a two modes here? So, if you look into this particular diagram, we have a two modes. Right? So, now when there are two modes, what is that Q of X is doing? Q of X is focusing only on one portion. So, when it is focusing only on one portion, other portion it is not at all capturing. So, this is a reverse KL. So, where it is you are punishing a Q for placing the probability where P has little or no mass when you consider a single mode.

[18:53] When it is a two modes or multi mode, then you are capturing only one mode. You are not at all focusing on the other mode. Right? So, with this, what is the drawback? We will see that. I hope forward and reverse is clear to you. Now, let us take one example and then understand what is this whole concept is all about. What is that given to you? For any problem statement, you will have a data set. Let us assume this is the data set that we have. And when you look into this particular data set, there are two features, feature one, feature two.

[19:32] What is your ultimate aim? Ultimate aim is to generate a new data. If you have to generate a new data, first you should learn the underlying data distribution. Now, uh visualization, now let us see how can you plot this and visualize its distribution. Now, this is the original data, feature one and feature two. » [snorts] » Now, original data distribution, we are denoting it as a P of X . What is P of X ? P of X is the probability distribution of this original data set. Now, how is that you have drawn this two graphs is it is an histogram. How is that you'll be drawing the histogram? Although I'm not getting into detail, but I'm giving an light idea. So, for the first histogram, whatever we have seen in a

[20:21] blue here, uh for the first histogram, we take all the values from the feature

one. So, from the feature one, take all of these values. Now, when you look into these values, there are some values which are you know, uh in a six range. There are some values which are in seven range. There are some values which are in eight, nine, 10, 11. There are so many different range range of values are there. So, what is that we take? We take all the values of this a feature one, divide their range into smaller intervals, and we call that a smaller intervals as bins. So, once we take those uh smaller intervals and divide that as a bin, then count how many values fall into each of these bin and x-axis gives the feature shows the

[21:14] feature one value range. And here this is the value range and y-axis whatever you are showing is number of samples in each range. What is the frequency [snorts] of those values? Number of such samples in each range is y-axis and in x-axis you are actually placing the values. Now, you can ask ma'am when we look into the data set, we don't have any values which are there in a range of four. No, it is a large data set. Instead of showing the entire data set, I have taken a small few set of samples and then placed here. Otherwise, the size of this data set is very large.

[21:58] I've taken only the screenshot of only few samples and those samples are placed on the left side. So, what is the your understanding for a given data set? For any problem statement that you solve, you will have a data set with you. For that data set, you want to know what is the distribution. And what is that how is that you'll be visualizing? Either you can go for an a discrete or you can go for a continuous. What we have plotted here is a discrete.

[22:24] How is that you have plotted the discrete one using an histogram. How is that you will plot an histogram? Take all the values. There are different range of values. Divide them into a separate intervals, we call it as bins. And count how many such values are there in that data set for that particular bin and then plot the graph. Now, whatever is the graph, similar way do it for feature two. Whatever are the two graphs that we have a drawn, these two are the original data set distribution. And that distribution we call it as p of x .

[23:00] And for a model we does doesn't know this a distribution. Model doesn't know that the graph is the perfect bell or there is a slight variation. The model doesn't know it. So, here the original distribution real data set of samples are drawn from some unknown true distribution and that distribution we call it as P of X . Now, what is VAE? VAE is all about the generative model, right? So, generative model main aim is to learn this distribution. How is that it is going to learn? It cannot learn the correct distribution. Why it cannot learn the correct distribution?

[23:39] Because we doesn't know. We for a model we are giving only few training samples. For a training with that training samples, it is going to learn some distribution. And let us assume that the distribution we call it as the Q of X . And what is the theta here? Definitely a neural network has to learn. If neural network has to learn,

it will learn through some parameters, that is weights and bias. That parameters we call it as the θ .

[24:06] Now, is it not we have two distributions? What is two distributions? One is a true P of X . Second distribution, the learned distribution Q of X . Now, what is the whole goal here? The goal here is Q of X should be as close as a P of X . This is the goal. If if the model can able to achieve this, then from this learned distribution, whatever the data that is generated, is it not the data will be closer to the original data or the true data? Agree?

[24:40] Yeah. So, this is what is the goal. The goal here is the whatever is the appropriate distribution the model is learning, that distribution should be closer to the original distribution. Now, you have to quantify if there is any mismatch between the learned distribution and original distribution, we have to quantify it. To quantify, we have KL divergence. That KL divergence, you can measure either forward or reverse. So, if it is a forward, it is P from Q , and this is a reverse KL divergence.

[25:14] Now, from the diagram, look at this. Now, this is the original blue is the original distribution P of X . Let us assume the model has a learned some distribution, and that learned distribution is a Q of X . And that is what is a shown in the dotted red diagram here. Look into this diagram. So, here the original P , whatever is the blue is the original distribution. Now, model has a learned some distribution. How did the model learn? Uh during the training process, it has seen some samples. From that the samples, it has learned some distribution. And towards the end of the training, the model has a learned some distribution Q of X .

[25:58] That learned distribution is denoted with dotted lines here. When you look into this a diagram very closely, uh you you can see here this a particular part. Look into this. Is it not there is a some distribution there, original distribution is there, but model failed to capture it. Can you see this? When you look into the X axis here, Q has not captured that. Q has almost reached to zero. So, is it not just by looking at this, we can say that model has not captured all the data properly.

[26:36] Now, which part of the data model has not captured? P has some value, but Q has not been covered there. So, measuring this is your forward KL. At the same time, look into the this particular what is the other scenario. The other scenario is probably the model would have learned something like this. You have a blue as original distribution and the learned distribution Q of X is that dotted red. When you look into this, observe very carefully what is that the model has a learned? The model has a learned something where P does not have the value at all. Look into this portion here. Here P does not have a value.

[27:19] P does not have a value here. But the model has a learned that part also. When model has a learned that part, is it not when it is a generating the data, it will generate some unrealistic data? Do you agree to this? I hope you have understood.

This is called reverse KL. So, whatever is the slide that you are seeing, it is for you to visualize with an example what is this P of X , what is this Q of X all about. So, this slide gives you the complete clarity about the KL divergence. I hope this is clear to you. Let me quickly summarize.

[27:55] Here we have the data set. The data set is already given to you. That data set has some probability distribution. That is called a real true probability distribution. That we are denoting it as P of X . Now, the whole idea here is the deep learning model or generative model should learn the distribution. Why it has to learn the distribution? Now, if you give that a distribution directly to the model, then model will generate exactly similar. Then there is no much variation. But if you allow the model to learn, it will not learn exactly. When it not learn exactly, then whatever it generates, it will be appropriate closer to the original one. Right? Then we are getting a different variants of that

[28:42] generated image. So, that is the reason why we don't tell the true distribution to the model. We allow the model to learn. When you are allowing the model to learn, it learns the approximation. Why it learns the approximation? Because it doesn't have the entire data set. From that few training samples, whatever it is learning, it will be an approximation. The goal is approximation should be closer to the original one.

[29:07] Now, how do you measure how far the model has learned? You will look into look into these two diagrams here. Uh the original is blue and the learned is red. So, here the learned model has failed to capture the data. Which data it has captured? Where P has a value but Q is not having a value. Measuring this is forward. In the second scenario, Q has captured extra data where P is not at all there. This is also wrong. So, this is all about your forward and reverse.

[29:40] Let us take one example and then understand this. Now, because we have seen something called, you know, it is capturing some additional data. It is not capturing the data. Let us take two examples, not one. This is the first example. And in this example, we are also learning two more concepts called mode covering and mode seeking. So, let us see what exactly is it these two. I have a data set. This data set is having a group of people. One set of people are having a height 160 cm. And the second set of people are having a height 190 cm. So, which means there are short people and there are tall people. And I have a very few people who are with the height 175.

[30:30] Uh I hope you have understood the problem statement. Now, for this data set, if you have to visualize this, is it not clearly you have two clusters? One cluster is where the samples belongs to one 60 cm. And there are other set of samples. The other set of cluster is where we have 190 cm tall. In between, there are very few little samples exist in between these two. In between these two is that 175. Okay.

Let us try to understand what is forward and reverse for this scenario. Now, when you look into forward KL, forward KL is also called as mode covering. What exactly is a forward KL? What is that it ensures? The first cluster, here cluster A is the all the points which belongs to 160 cm. And the cluster B here is all

[31:24] those samples of 190. So, these two are the two clusters. What is forward KL divergence ensure? It ensures that that Q has to spread and cover both the modes. This is what the forward KL says. So, a single Gaussian Q of X becomes so broad so that it covers both the regions. Now, if this is the case, what is going to happen? Can you imagine if you are going to spread Q so that it has to cover both the regions? Because your aim is it has to cover both. That is where it is called mode covering. So, both the groups are represented. Very good. But, what exactly is the limitation here?

[32:12] Look at the Q of X very carefully and look at where it has, you know, the maximum probability. The maximum probability is here. It assigns the probability to this 175. Which means even for a low density region, low density region meaning we have very few samples of 175. But, because you spreading that Q very broadly with the intention that it has to cover both, it is going to generate the samples which are more of that 175 cm. But ideally, we have very less such samples. We want the data which is either from 160 or we want the data from 190. But what is forward KL divergence says, cover both. In the process of covering both, it is generating the samples which are intermediate. Therefore, it is going

[33:10] to give some unrealistic data. This is all about forward KL. Let us look into So, what is forward KL? It may generate the same samples around 175. Even though there are very such limited samples which are there in the examples. Now, let us look into what is the reverse KL. I hope forward is clear to you. Now, let us see what is the reverse KL. Go back and then visualize You will have two graphs and those graphs are that particular graph is something like this. You have a cluster A and then we have a cluster B.

[33:47] And the cluster A is all about 160 and the cluster B is all about 190. What is the reverse KL is the learned distribution will mainly focus only on one mode only. So, now if it is focusing on a cluster A, then whatever the images that it is going to generate or whatever the data that it is going to generate, it is going to generate the data which belongs to cluster A. Suppose, if the model is covering this part. If the model is a covering of this part, then it is going to generate only those samples which belongs to cluster B.

[34:29] Which means it is going to generate the samples who is very whose parameters are very belongs to a tall people. So, what exactly is a reverse KL problem? It is going to a single Gaussian Q of X may concentrate on either 160 mode or 190 mode. So, what is an advantage? Advantage is the samples from the selected mode are very realistic. Now, if the model has a learned to Q of X, that Q of X is closely matching

with the only the probability distribution of the cluster A. Now, from this distribution when you generate the samples, what is that you're going to get? You're going to get the data which are more closer to this cluster A distribution, right?

[35:17] Now, what is one disadvantage of this? The disadvantage is the other mode is not at all covered, which means you will not be able to generate the data from that other mode only. So, reverse KL ignores the other valid group, but whatever the samples it is a generating, those samples are from that chosen group. If at all if it is covering this, whatever the samples it is generating, from that samples it will be realistic and it will be closer to that particular cluster. So, this is all about one example what we have given.

[35:55] And let us take the last example so that the visualization is more clear to you. Again, we are discussing forward and the reverse only, but now I'm showing you with an example how it is going to have an unrealistic data. Consider we have a data set where you have a cats and dogs images. So, you have a set of cats images and set of a dogs images. This is the data set that we have. Now, what we are doing? What is our task? Our task is to generate animal images from this particular data set. Which means whatever the images that you're going to generate, it will be a different version of cat or the dog because the training examples are cats and dogs here.

[36:39] Now, if it is an forward KL, what is a forward KL? Always the aim is it has to cover both the modes. So, which means it has to cover both the cats and dogs. If it has to cover both the cats and dogs, so definitely it is going to How is that your diagram? Go back and then visualize the diagram. You have both these distributions and then the learned distribution will be something the broader. So, because of this, it is going to generate an unrealistic data.

[37:11] How is that unrealistic data? Can you look into this? Here you have a cats, but whatever you are getting is very unrealistic one. It looks It is a combination of both cat and a dog. So, if you look into this is cat images, but when you by the time you come here, it is the features are more towards the dog image. So, forward KL divergence, because your aim is it has to cover both the modes. In this process, you are spreading Q. When you spread the Q, it covers both the P both the clusters.

[37:48] When it covers both the clusters, the majority is at the center. So, when the majority is at the center, where you don't have less samples in the previous example. In this, when the majority is at the center, it is going to combine both and then it is going to give us the data. In this process, no doubt it is going to generate the data, but those data will be unrealistic data. So, mode covering covers all the modes, but may generate an unrealistic in between samples, right? Let us look into what is the reverse KL. Reverse KL mainly aims that if there are multiple modes, I have to focus on learning one mode properly. So, either it will learn a cat properly or it will learn a dog properly. But, when it generates an

[38:36] image, it will generate only what it has a chosen. So, if you look into this, here it is it has a captured or it will capture only one mode. It has a generated the cat images, but dog images are missing. Here, it has generated a dog, but cat images are missing. It has generated a cat, dog images are missing. So, this is called mode seeking. What is a seeking? It generates a sharp realistic samples, but only from one mode. The other mode is completely missing. So, it is mode seeking.

[39:10] Um so, this is all about this particular session. Although we have not completed KL divergence, we have only seen some of the fundamental example, you know, concepts, definitions of KL divergence. So, in the upcoming session, we will look into the most important concept of KL divergence called self-information, entropy, cross-entropy, and then we will see the detailed derivation of KL divergence. Thank you.